

SEDAR 31: Gulf of Mexico Red Snapper

Data Workshop

Life History

- Review the evidence for density dependence in older ages (e.g. ages 2-3). Incorporate full age model of recruitment to examine density-dependent effect.
- Site and habitat specific comparisons from more regions of the Gulf are needed for estimation of age-0 and age-1 mortality, accounting for shelf characteristics (e.g., width, slope, depth) in tests of density-dependent variation in M and emigration.
- Broader understanding of habitat value and areal estimates of habitat (distribution—areas of trawlable vs. untrawlable bottom; more refined maps Gulf-wide etc) are needed to further inform the habitat limitation hypothesis for density dependence.
- Assess the impact of potential predation/competition for taxa of particular interest (lionfish, marine catfish, sciaenids, and red grouper). As well, investigate alternative population regulatory mechanisms including potential sources of density-independent increases in mortality and distant sources of recruitment (but see stock delineation section).
- Evaluate the potential for sea-bottom restoration or other means to expand habitat and increase survival for post-settlement red snapper.
- The LHW recommended that existing otolith archives (e.g., NOAA) be used to further investigate interpretation of increment formation based on section orientation, sample source (location), season and year. This could be conducted as a graduate student project in collaboration with agency personnel.
- Interested Academic representatives (e.g. Auburn University, University of S. Florida) should be included at Gulf States Marine Fisheries Commission sponsored otolith workshops (e.g., May 2013) to review age determinations and promote standardization.
- Based upon the results of Szedlmayer and Beyer (2011 SEDAR31-RD20), further investigation of longevity is warranted. More recent catches of older fish should allow a direct comparison to ¹⁴C coral chronologies established during the nuclear testing period and extend the age that can be directly validated (beyond 38 years in the earlier study by Baker and Wilson 2001).
- A general recommendation of the LHW is to expand design-based fishery-independent sampling to elucidate regional (i.e., eastern and western GOM) and sub-regional differences in the demographics of red snapper.
- A further recommendation is to increase random, representative sampling of the catch in order to avoid clustering effects and non-representative sampling which could lead to spurious differences in growth rates. Alternatively, and for localized- or small-scale studies, corrections for length limits and appropriate weighting may need to be utilized to treat data gaps, missing ages and adjust for selectivity (see Chih 2012 SEDAR31-DW18).
- Future surveys should collect ovarian samples fixed in formalin for histology analysis, spawning marker fraction analysis and age/size at maturity analysis.
- Additional fecundity collections are necessary from all areas of the Gulf.

- Additional research is necessary to further clarify regional reproductive and demographic differences.
- More information is needed to understand movement of young and older adult red snapper across along shore barriers. In particular the LHW recommends a large scale tagging study focused west and east of the Mississippi River.
- Telemetry versus tagging approaches need to be expanded and evaluated according to shelf characteristics; e.g. cross compared in areas with little natural hard bottom habitat (yet high artificial reefs) versus areas with relatively high areal coverage of hard bottom and with more dispersed artificial reefs.
- The LHW recommends a workshop or research symposia be convened to synthesize results and assess methodology for estimating red snapper movements and home range.
- In order to reduce measurement error in the future, the LHW recommends that port agents, observers and field scientists record maximum total length for red snapper.

Commercial Fishery Statistics

Landings

- Revisit how the historical landings were constructed.
- Explore ways to ensure that IFQ and trip ticket landings match.
- Apportion landings accordingly in ALS for TX landings with missing gear.

Discards

- Add species to discard logbook form.
- Provide better instructions on how to complete the discard logbook.
- Consider and use relevant input from external review of observer program.
- Social and economic impacts on fisher behavior in terms of fish discards.
- Better determine available allocation to vessels on a given trip.

Length/Age

- Standardize length and age data formats from various data sources.
- Build age databases with Trip ID number for FL and FIN data.
- Evaluate how to handle catch at age of non-representative age samples.

Recreational Fishery Statistics

- Evaluate the technique used to apply sample weights to landings. Investigate the SEFSC
- Method by analyzing the order of variables in the hierarchy and the minimum number of fish used. Furthermore, evaluate alternative methods, including a meta-analysis of the existing information from difference sources, areas, states, surveys, etc. that could be performed.
- Develop methods to identify angler preference and targeted effort. Require a reef fish stamp for anglers targeting reef fish, pelagic stamp for migratory species, and deep-water complex stamp

for deep-water species. The program would be similar to the federal duck stamp required of hunters and could help managers identify what anglers were fishing for.

- Continue and expand fishery dependent at sea observer surveys to collect discard information. This would help to validate self-reported headboat discard rates.
- Track Texas commercial and recreational discards.
- Estimate variances associated with the headboat program.
- Evaluate existing and new methods to estimate historical landings. Hind-casting of red snapper landings is complicated by a lack of reliable historical effort data. To get at estimating historical effort, analysts could track consumables (gas, ice, bait) to develop price indices.
- Investigate how CPUE changes over time due to technological advances and changes in fishing practices.

Measures of Population Abundance

The following are research recommendations that may improve the utility (precision) of the

SEAMAP larval index for red snapper.

- Expand the use of molecular genetics to identify the smallest and most abundant snapper larvae in SEAMAP samples that cannot be positively identified as red snapper because diagnostic morphological characters are not yet developed.
- Begin directed sampling for fish eggs on SEAMAP summer trawl and fall plankton surveys using vertical nets hauls. The protocols for fish egg sampling have been established by NMFS/SWFSC scientists and are in use on the west coast. Fish egg collections are easy to make and take little additional sampling time. The eggs in these samples would have to be identified genetically but the protocols for genetic identification of red snapper eggs have been worked out by Frank Hernandez and Keith Bayha at DISL. The results of their MARFIN funded project using CUFES samples from our SEAMAP Fall Plankton surveys are impressive and significant. They produced maps of red snapper egg (i.e. spawning) distribution over the entire Gulfwide survey area.
- Estimates of egg abundance data coupled with the updated reproduction parameters spawning frequency and fecundity) generated by NMFS scientists at the Panama City Lab could eventually be used to produce an actual spawning biomass estimate for red snapper.
- Continue aging red snapper larvae from SEAMAP samples to improve the age-length relationship (key). This should improve the precision of the SEAMAP larval abundance index that is now based on a single age class of larvae across years.
- Produce a SEAMAP larval index based on the abundance of red snapper larvae captured during SEAMAP summer shrimp/groundfish surveys (past and present). This survey has for a number of years now been expanded to include the entire northern Gulf of Mexico shelf. The data from summer months (i.e. during peak red snapper spawning months) could be a far better indication of spawning production than data from the end of season from which the current SEAMAP larval index is derived.
- Explore the utility of a larval red snapper index based on a comprehensive modeling approach that includes all SEAMAP stations (regardless of how many times they have been sampled over

the time series) and both sampling gears, i.e. neuston and bongo samples. There are other likely explanatory variables (e.g., salinity) that could ultimately improve the index.

Discard Mortality Rate

Future surveys, at minimum, should be structured around quarterly sampling, collect water temperature profile data, reflect the range of depths associated with the fishery, and strive to calculate season and depth specific estimates. Due to the limited number of experiments evaluating the relationship between thermal stress and release mortality, it is strongly encouraged that investigators measure and report water temperatures and thermocline profiles associated with capture. More studies evaluating the use of bottom release devices are also needed. Future discard observation surveys should collect frequency data regarding specific barotraumas incurred and loss of reflex response, because similar relationships could be developed as better techniques are developed to measure the delayed mortality component. Experiments estimating impairment scaling, and both immediate and delayed mortality, would be particularly useful so that a relationship among components could be developed and historical immediate release mortality estimates could potentially be adjusted.

Assessment Workshop

No recommendations were provided.

Review Workshop

Below, the RW Panel highlights research recommendations they feel should be emphasized, and provides new recommendations partly based on assessment methodology and results.

Age and Mortality

The RW Panel recommends that research effort be focused on the issue of ageing error, both within and among ageing facilities. A more comprehensive analysis of ageing error should permit its inclusion in the SS3 model.

There appeared to be some confusion in the DW report as to the purpose of and resultant data from bomb radiocarbon analysis of otoliths. This method is a means to evaluate the estimated birthdate of a fish relative to the $\Delta^{14}\text{C}$ preserved in other aragonitic structures, such as corals. Radioactive ^{14}C was enriched in oceanic waters in the late 1950s and early 1960s following above ground nuclear weapons testing. Coral skeletons, for example, reflect this enrichment by having peak $\Delta^{14}\text{C}$ values in skeletons formed during the early to mid 1960s and then declines thereafter. If opaque zones in otoliths are formed annually, then fish estimated to have birthdates in the early 1960s should have similar high $\Delta^{14}\text{C}$ values at the core of their otoliths. Other radio chemistry validation techniques, such as $^{210}\text{Pb}/^{226}\text{Ra}$ dating, provide estimates of absolute fish age; bomb radiocarbon analysis only provides a relative age estimate but can be used to validate opaque zone formation. Both of these age validation techniques have been applied to red snapper, along with other validation and verification techniques. In fact, no other marine fish has been the subject of as many different age validation/verification studies as red snapper. Results of these studies are overwhelming: opaque zones in red snapper otoliths are formed annually.

Growth

The RW Panel recommends further analysis on the growth function fit to size at age data from 2003-11. The fitted model included in the assessment tends to overestimate size at age for fish <5 yr, overestimate size at age for fish 5-10 yr, and underestimate size at age for fish >20 yr. Part of this results from the manner in which the model accounts for variable size limits through time. However, the RW Panel expressed concern whether some of the observed variability in size at age in the data resulted from ageing error between laboratories. In the future, modeling growth with a random effects approach may be more appropriate.

Population Structure

The RW Panel reiterates various research recommendations focused on the population structure of Gulf red snapper. Hydrographic models should continue to be employed to estimate potential larval dispersal within the US Gulf, between the eastern and western Gulf, and on smaller spatial scales. A large-scale conventional tagging study might be useful to examine post-settlement mixing both between the eastern and western Gulf and within these areas. Lastly, advances in restriction-site-associated-DNA (RAD) sequencing mean that much more powerful genetic population structure analysis is now possible relative to historical mitochondrial DNA or microsatellite DNA approaches previously applied to Gulf red snapper.

Discard Mortality

Estimation of dead discards is a product of the number of discards and the discard mortality rate, both of which are highly uncertain for red snapper. Observer data in the shrimp trawl and directed commercial fisheries enable estimates of the magnitude of discards. There are much more limited data available in the recreational fishery to estimate the magnitude of discards. There are some observer-based estimates available for the headboat and charter boat sectors, but efforts to collect those data should be expanded. Reliance on self-reported discards in the MRIP to estimate discards in the private recreational sector is problematic with no clear solution. Electronic reporting through smart-phone applications does provide for instantaneous reporting of discards, but the process involved remains reliant on self-reporting which has been shown to be biased in other sectors where both self-reporting and observer-based estimates of discards are available.

Further research appears warranted with respect to estimating the magnitude of discards among fishery sectors, as well as providing more robust estimates of post-release mortality. Few of the existing discard mortality studies address the issue of depredation on released fish and that should be a focus moving forward. Several existing lines of research indicate chronic effects of barotrauma which may lead to mortality are pervasive in released red snapper, and that studies which simply examine surface condition or submergence of released fish may grossly underestimate release mortality. Therefore, a focus moving forward should be on conducting studies that examine both depredation on released fish and chronic versus acute mortality caused by catch and release.

Episodic Mortality Events

Episodic events have the potential to impact red snapper population ecology in the northern Gulf of Mexico. Among recent and ongoing events that have this potential are hypoxia associated with plumes of the Mississippi and other northern Gulf rivers, harmful algal blooms, particularly along the west Florida shelf, and the Deepwater Horizon Oil Spill (DHOS). Potential impacts of the DHOS were discussed during the RW but little work had been done attempting to examine potential impacts in either the DW or AW. In fact the words “Deepwater Horizon” appear only once in the DW Report and never in the AW Report. Part of this issue may stem from the fact that if potential impacts were restricted to recruitment effects then an assessment model would not capture that signal until affected cohorts moved into the fishery. Future assessments of Gulf red snapper should be conducted with the explicit goal of attempting to model any enduring DHOS effects.

Recommendation for Research on Improvement of the SS Model

The RW Panel recommends changing the model’s code to enable separate sets of SR parameters to be estimated for the different population subunits.

Recommendations for Improvement of the SEDAR Process

The third charge included in RW TOR number 6 is to provide recommendations on possible ways to improve the SEDAR process. Improvements should include:

- The most critical need is for timeliness in completion of tasks and reports. The SEDAR process is complex and demanding, involving scientists with diverse areas of specialization and including a large array of issues and concerns. Completion of work requirements on schedule is challenging and demanding, but the more closely that deadlines are met, the more efficient and productive the process can be.
- More standardization of report format would be helpful. Sections of reports are written by different individuals and groups, all of which have their own writing styles and preferences, but content of reports would be improved if each workgroup provided summaries of their results and conclusions, enumerated or in paragraph form. In addition, a more uniform identification of procedural and research issues, presented at the end of each workgroup section would be informative. Proposals and rationale for further study have potential for moving forward directly on problems that are recognized as especially important.
- Given that the AW analyzes the extant databases for the species under consideration, the group would be well placed to be critically aware of additional data needs. Recommendations for future research could profitably be a standard part of their SEDAR report.

Guidance on key improvements in data and modeling approaches which should be considered when scheduling the next assessment:

The RW Panel expressed serious concerns regarding the amount of time allotted for review of this assessment. As noted above, the AW Report was provided to the RW Panel on Friday, April 26th for a

RW beginning on Monday, April 29th. Furthermore, the AW Report had not been reviewed by the AW Panel. The AW Report was incomplete, contained errors, and the documentation of the model inadequate for a thorough review.

The RW Panel recommends that given the data and model complexities inherently associated with stock assessment of Gulf red snapper, more realistic timelines be considered for the next assessment.